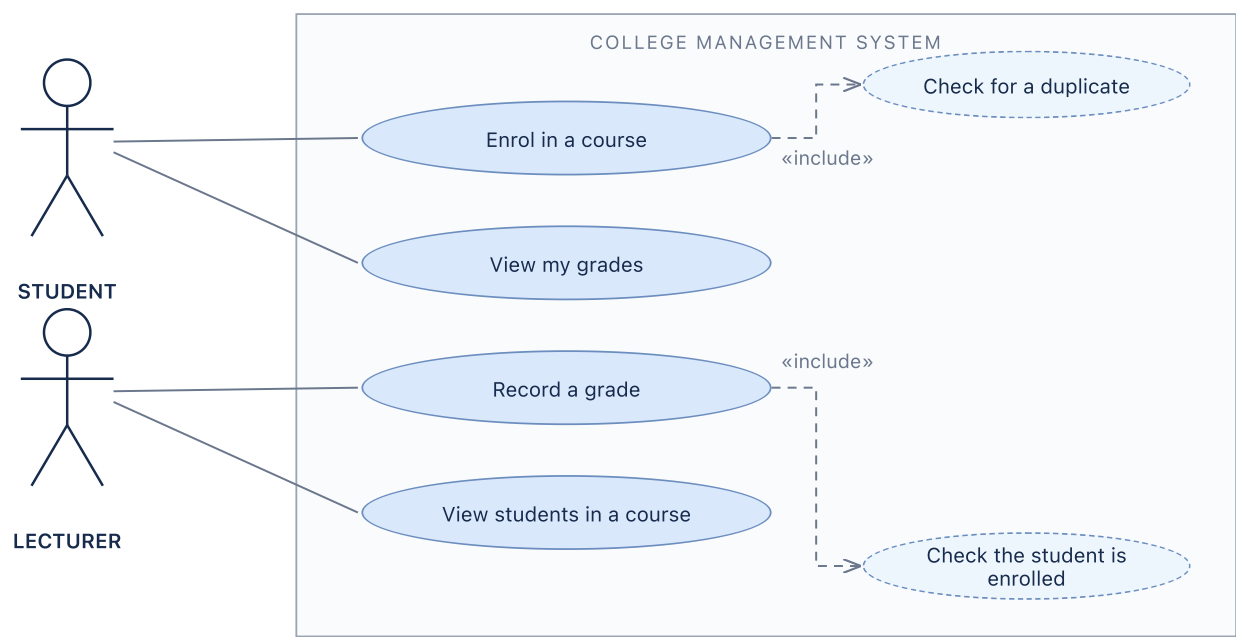


Use Case, Activity and Class Diagrams

1. Use Case Diagram

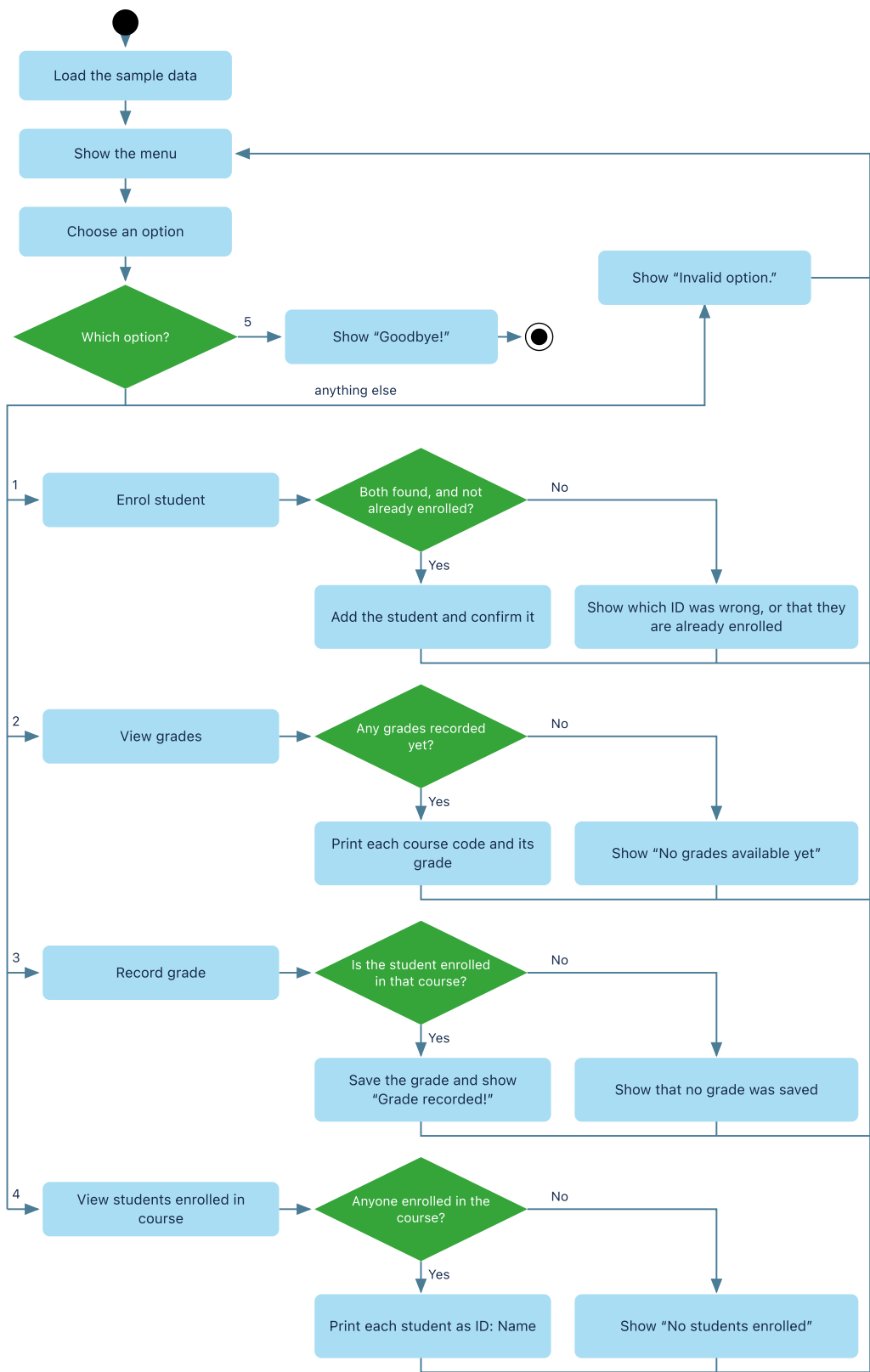
Two actors: A student enrolls in a course and looks at their own grades. A lecturer records a grade and lists who is in one of their courses.

These two carry a check that has to pass first. Enrolling checks the student is not already on the course, else returns false rather than adding them twice. Recording a grade checks the student is enrolled in that course, and shows error message if they are not.



2. Activity Diagram

Everything the program does, from startup to exit. The sample data is loaded once, then the menu repeats until option 5 is chosen. Each of the four working options is a row: look up the IDs that were typed in, ask one question, and either do the thing or explain why it cannot be done. Anything outside 1 to 5 is turned away and the menu comes back.



3. Class Diagram

Five classes. **University** is the container: three dictionaries keyed by ID, and the three add methods that fill them. **Config** sits outside the model and only use: it loads the sample data and prints the menu, so it is drawn as a dependency rather than a part of the structure.

The three interesting links are between the other three. A lecturer teaches many courses, and `teach()` sets both sides at once, so a course also knows its lecturer. Students and courses are many-to-many: a course keeps a list of its students, and each student keeps their grades in a dictionary keyed by course code. There is no separate enrollment class here.

